

EE150 Signals and Systems Final Exam 2025 Spring

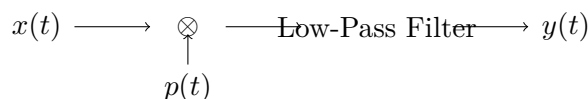
魏之远回忆版

Problem 1

Given unit impulse response of a LTI system $h[n] = \frac{\sin(\frac{\pi}{2}n)}{\pi n}$

- a) Find $H(e^{j\omega})$
- b) Compute DTFT of input signal $x[n] = \cos(\frac{\pi}{8}n) + \sin(\frac{3\pi}{4}n)$
- c) Find output signal $y[n]$ when input is $\uparrow$ (unit impulse)

Problem 2



As shown above, the sampling system receives a continuous-time input signal $x(t) = 2T^2 \cos\left(\frac{2\pi}{T_1}t\right)$.

$T_1 = 1ms$, $p(t)$ is given by $p(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_s)$.

$T_s = 0.2ms$, $\delta(t)$ is unit impulse function.

- a) Find $X_s(j\omega)$ of signal $X_s(t) = p(t)x(t)$.
- b) Sketch the spectrum $X_s(j\omega)$ of $x_s(t)$.
- c) Determine the range of cutoff f for the ideal low-pass filter.

Problem 3

Consider a causal & stable continuous LTI system that can be described by $\frac{d^2y(t)}{dt^2} + 9\frac{dy(t)}{dt} + 20y(t) = \frac{dx(t)}{dt}$, where $x(t)$, $y(t)$ are input & output.

- a) Determine $H(s)$
- b) Plot pole-zero pattern of the system, determine ROC
- c) Determine system output $y(t)$ if $x(t) = 10u(t)$.

Problem 4

Consider a discrete-time LTI system: $y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = x[n] + \frac{1}{3}x[n-1]$

- a) Determine $H(z)$. Plot pole-zero pattern.
- b) If the system is causal, determine ROC & unit impulse response.
- c) ... stable ...
- d) Plot block diagrams in both direct & parallel form.